

The Protonmotive Q Cycle

ENERGY TRANSDUCTION BY COUPLING OF PROTON TRANSLOCATION TO ELECTRON TRANSFER BY THE CYTOCHROME *bc*₁ COMPLEX*

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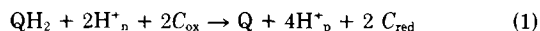
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The cytochrome *bc*₁ complex is an oligomeric membrane protein complex which is a component of the mitochondrial respiratory chain and of the electron transfer chains of numerous bacteria which use oxygen, nitrogen, and sulfur compounds as terminal electron acceptors. The cytochrome *bc*₁ complex also participates in the cyclic transfer of electrons to and from the photosynthetic reaction centers in anoxygenic photosynthetic bacteria.

In all of these species the cytochrome *bc*₁ complex transfers electrons from ubiquinol to cytochrome *c* and links this electron transfer to translocation of protons across the membrane in which the *bc*₁ complex resides. The mechanism by which the cytochrome *bc*₁ complex links electron transfer to proton translocation is the protonmotive Q cycle (1). This protonmotive electron transfer is one of the most important mechanisms of cellular energy transduction, found in a phylogenetically diverse range of organisms (2). The purpose of this review is to explain the protonmotive Q cycle.

The Cytochrome *bc*₁ Complex

The cytochrome *bc*₁ complex transfers two electrons from ubiquinol (QH₂) to two molecules of cytochrome *c*, deposits two protons from ubiquinol on the positive side of the membrane, and translocates an additional two protons from the negative to the positive side of the membrane per pair of electrons transferred. The reaction catalyzed by the *bc*₁ complex is described by Equation 1, where the subscripts *n* and *p* designate negative and positive sides of the membrane, respectively, and C_{ox} and C_{red} refer to oxidized and reduced cytochrome.



This protonmotive activity of the cytochrome *bc*₁ complex converts the free energy of the electron transfer reaction into a proton gradient across the membrane. In mitochondria the *bc*₁ complex is located in the inner mitochondrial membrane, and the cytoplasmic side of the inner membrane is positive relative to the matrix. In bacteria the *bc*₁ complex is located in the plasma membrane, and the periplasmic side of the membrane is positive relative to the cytoplasm.

The cytochrome *bc*₁ complex contains three electron transfer proteins; cytochrome *b*, which contains two heme *b* groups non-covalently bound through bis histidine residues (3, 4), cytochrome *c*₁, which contains a heme *c* group covalently attached to the protein by thiol ether linkages to cysteines (5), and an iron-sulfur protein, which contains a 2Fe:2S cluster non-covalently liganded to the protein through two

cysteines and two histidines (6). Cytochrome *bc*₁ complexes purified from bacteria such as *Paracoccus denitrificans* (7) and *Rhodospirillum rubrum* (8) consist of only the three electron transfer proteins. The cytochrome *bc*₁ complexes of mitochondria contain numerous polypeptides, lacking redox prosthetic groups, in addition to the three electron transfer proteins (2). However, a comparison of the *bc*₁ complexes of *P. denitrificans* and bovine heart mitochondria showed that the protonmotive Q cycle mechanism is operative in both bacteria and mitochondria (9).

To illustrate structure-function relationships between the proteins of the cytochrome *bc*₁ complex and the protonmotive Q cycle, I have depicted in Fig. 1 the three electron transfer proteins essential to the Q cycle mechanism. Cytochrome *b* is a transmembrane protein (10, 11). The two heme groups of cytochrome *b* form an electrical circuit across the mitochondrial or bacterial membrane, and an applied membrane potential moves an electron from one heme to the other (12).

The potential of the *b*-560 heme is approximately 60–120 mV more positive than that of *b*-566 in all cytochrome *bc*₁ complexes thus far characterized. This increment in potentials facilitates electron transfer from the *b*-566 to the *b*-560 heme in the face of an opposing membrane potential (see Fig. 1).

Cytochrome *c*₁ and the iron-sulfur protein are located on the electropositive, P side, of the inner mitochondrial membrane or bacterial plasma membrane (2, 13). Cytochrome *c*₁ is imbedded in the *bc*₁ complex with the heme-containing portion of the protein located at the surface of the membrane (14). The iron-sulfur protein is held to the surface of the *bc*₁ complex by hydrophobic interactions (15) and can be reversibly removed from the complex by extraction with detergent (16). The ability to resolve and reconstitute the iron-sulfur protein provided a methodology to document the protonmotive Q cycle pathway (17, 18).

The Protonmotive Q Cycle Mechanism

The protonmotive Q cycle, shown in Fig. 2, describes the pathway of electron transfer among the redox prosthetic groups of the cytochrome *bc*₁ complex and accounts for linkage of proton translocation to this electron transfer. In step 1 of the Q cycle ubiquinol (QH₂) is oxidized at center P in a concerted reaction in which one electron is transferred to iron-sulfur protein (ISP) to form a ubisemiquinone anion (Q_p^{•-}), which immediately reduces the *b*-566 heme. This oxidation deposits two protons on the positive side of the membrane. One is released coincident with oxidation of ubiquinol to ubisemiquinone; the second is released essentially simultaneously with the first as ubisemiquinone ionizes to ubisemiquinone anion.¹

The two electrons from ubiquinol thus diverge at center P. One is transferred to iron-sulfur protein. In step 2 this electron is transferred to cytochrome *c*₁ and then to cytochrome *c*. The second electron from ubiquinol, transferred from Q_p^{•-} to *b*-566, recycles through the *bc*₁ complex as it is transferred from the low potential *b*-566 to the higher potential *b*-560

¹ Although ubisemiquinone is ionized at physiological pH, the ionization state of ubisemiquinone at center P does not affect the proton translocation stoichiometry. If ubisemiquinone is protonated at center P, the release of the two protons from ubiquinol is temporally, but not spatially, separated.

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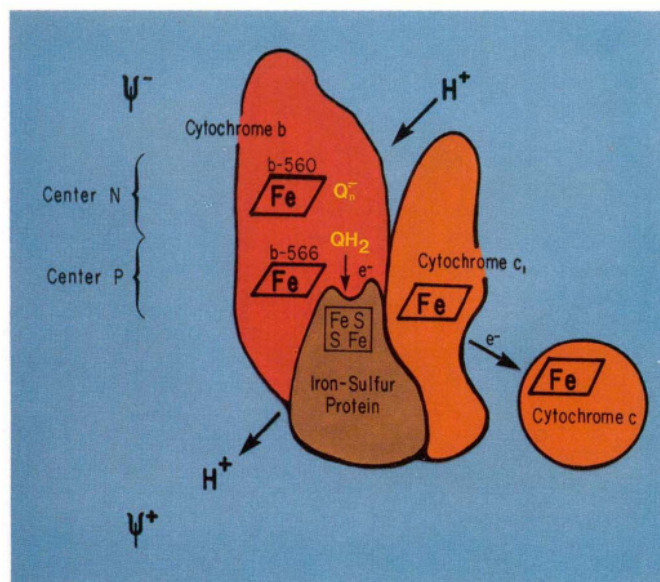


FIG. 1. Topographical arrangement of the three electron transfer proteins of the cytochrome bc_1 complex. The complex is asymmetric and spans the inner mitochondrial membrane or bacterial plasma membrane, with cytochrome c_1 and iron-sulfur protein located on the electropositive, P, side of the membrane. Cytochrome b is transmembranous, with the b -566 heme located near the P surface of the membrane and the b -560 heme located toward the electronegative, N, side of the membrane. The ubiquinol (QH_2) oxidation site, referred to as center P, is at an interface between the iron-sulfur protein and cytochrome b , proximal to the b -566 heme. A stable semiquinone radical ($Q^{\bullet-}$) is shown at center N, proximal to the b -560 heme, where cytochrome b alternately reduces ubiquinone to ubisemiquinone and ubisemiquinone to ubiquinol. The diagram also shows proton uptake and release at centers N and P, respectively, and electron transfer from cytochrome c_1 to cytochrome c . The stoichiometries of proton translocation are discussed in the text.

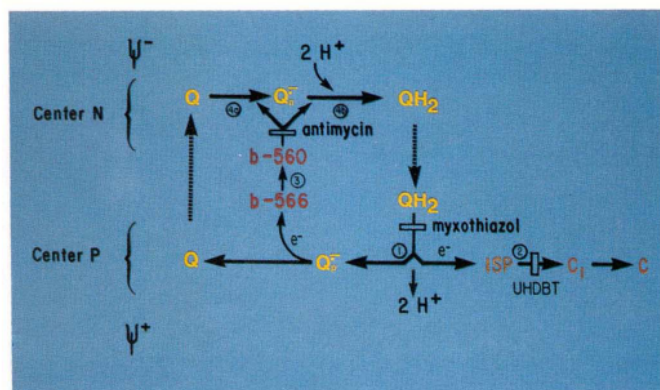


FIG. 2. Protonmotive Q cycle mechanism of electron transfer and proton translocation in the cytochrome bc_1 complex. The scheme shows the branched cyclic pathway of electron transfer from ubiquinol (QH_2) to cytochrome c through the four redox centers of the cytochrome bc_1 complex. The circled numbers designate electron transfer reactions. In step 1 ubiquinol is oxidized at center P in an essentially concerted reaction in which one electron is transferred from ubiquinol to iron-sulfur protein (ISP), generating a low potential ubisemiquinone anion ($Q^{\bullet-}$), which immediately reduces the b -566 heme group. Two protons are released at the P surface of the membrane coincident with oxidation of ubiquinol to ubisemiquinone anion.¹ In step 2 the electron transferred to iron-sulfur protein is transferred to cytochrome c_1 and then to cytochrome c . In step 3 the electron transferred to the b -566 heme group is transferred against the membrane potential inward through the membrane from the b -566 to the b -560 heme (see Fig. 1). Steps 2 and 3 can, and probably do, occur simultaneously. In step 4a the b -560 heme reduces ubiquinone to the relatively stable ubisemiquinone anion ($Q^{\bullet-}$). When b -560 is re-reduced by a repeat of the above series of reactions, b -560 reduces ubisemiquinone anion to ubiquinol (step 4b). Reduction of ubisemiquinone anion to ubiquinol at center N consumes two protons at the N surface of the membrane. The divergent oxidation of ubiquinol at center P (step 1) occurs twice during one complete Q cycle. One complete Q cycle thus deposits four protons on the positive side of the membrane, reduces two cytochrome c molecules, and consumes two protons from the negative side of the membrane. The open boxes show the sites at which myxothiazol, UHDBT, and antimycin inhibit electron transfer reactions within the complex. The effects of these inhibitors on pre-steady state reduction of the cytochromes and how they have been used to substantiate the Q cycle mechanism are described in the text.

(Fig. 2, step 3). Cytochrome b -560 then reduces ubiquinone (Q) to ubisemiquinone anion ($Q^{\bullet-}$) at center N (Fig. 2, step 4a).

At this point the Q cycle is only half complete, since only one electron from ubiquinol has been transferred to cytochrome c . This "half-cycle" deposits two protons on the P side of the membrane, while the second electron from ubiquinol transiently resides on the ubisemiquinone anion at center N ($Q^{\bullet-}$). As discussed below, $Q^{\bullet-}$ is relatively stable.

A second molecule of ubiquinol is then oxidized by iron-sulfur protein, transferring one electron to cytochrome c_1 en route to reduce a second molecule of cytochrome c , again forming the ubisemiquinone anion, $Q^{\bullet-}$, and depositing two more protons on the P side of the membrane. $Q^{\bullet-}$ reacts by the same pathway as previously, transferring an electron to b -566 and b -560. Cytochrome b -560 then reduces the previously formed stable ubisemiquinone anion ($Q^{\bullet-}$) to ubiquinol at center N, consuming two protons from the negative side of the membrane and completing the Q cycle (Fig. 2, step 4b).

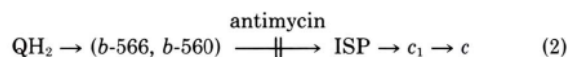
During one complete Q cycle two molecules of ubiquinol are oxidized to ubiquinones, but one molecule of ubiquinol is regenerated by rereduction of one of these ubiquinones. One complete Q cycle requires that the iron-sulfur protein, cytochrome c_1 , and the two hemes of cytochrome b undergo two "redox turnovers," duplicating steps 1, 2, and 3. In the course of one Q cycle cytochrome b -560 alternately reduces ubiquinone to ubisemiquinone anion (step 4a) and ubisemiquinone anion to ubiquinol (step 4b).

As a result of the net oxidation of one ubiquinol, two molecules of cytochrome c are reduced, four protons are deposited on the positive side of the membrane coincident with the two divergent center P oxidations, and two protons are consumed on the negative side of the membrane as b -560 rereduces a ubiquinone to ubiquinol. This stoichiometry of proton translocation implies that for each one molecule of cytochrome c reduced, a pH meter will detect deposition of two protons outside of the mitochondrial membrane, while measurements of potassium uptake in the presence of valinomycin will detect only one compensating positive charge electrophoresed through the membrane. These stoichiometries have been repeatedly obtained with mitochondria (19, 20), and purified cytochrome bc_1 complexes reconstituted into liposomes (9, 21).

Experimental Evidence for the Protonmotive Q Cycle

The protonmotive Q cycle accounts for the stoichiometry of proton translocation in the bc_1 complex and, as explained below, uniquely accounts for how cytochrome b -566, which has an oxidation-reduction potential much more negative than that of the ubiquinone/ubiquinol couple, can participate in oxidation of ubiquinol. The Q cycle also accounts for one of the most enigmatic observations relating to cytochrome b oxidation and reduction, a phenomenon known as "oxidant-induced reduction of cytochrome b ."

To appreciate the unique manner in which the Q cycle describes oxidant-induced reduction of cytochrome b and to illustrate the shortcomings of linear schemes of electron transfer, it is useful to consider a linear sequence of the type shown in Equation 2.



In this and similar linear schemes, the electron transfer components are arranged in descending order according to their oxidation-reduction potentials, although the placement of b -566 in such a linear sequence is problematic due to its

low oxidation-reduction potential. The linear scheme also includes a site for action of antimycin, a respiratory inhibitor that causes electrons to accumulate in the *b* cytochromes and ubiquinol, while allowing the *c* cytochromes and iron-sulfur protein to be oxidized (22).

In such a linear scheme one would anticipate that a sudden increase in the rate of cytochrome *c* oxidation would be followed sequentially by increased oxidation of cytochrome *c*₁, iron-sulfur protein, the *b* cytochromes, and ubiquinol. However, Chance discovered that while addition of a pulse of oxygen to slowly respiring mitochondria caused the expected increase in oxidation of the cytochromes *c* and *c*₁, this was accompanied by a transient increased reduction of the *b* cytochromes. This oxidant-induced reduction of cytochrome *b* involves both *b* cytochromes, and the extent of *b* reduction is magnified by the addition of antimycin (23). No linear scheme of electron transfer accounts for oxidant-induced reduction of cytochrome *b*.

Wikström and Berden (24) correctly explained the oxidant-induced reduction of cytochrome *b* by postulating that ubiquinol is divergently oxidized by cytochrome *c*₁ and cytochrome *b*. Mitchell (1, 25) conceived the protonmotive Q cycle and incorporated the divergent oxidation of ubiquinol. In the Q cycle, oxidation of cytochrome *c* causes sequential oxidation of cytochrome *c*₁ and iron-sulfur protein (Fig. 2). The latter in turn oxidizes ubiquinol and generates a reductant, Q_p^{•-}, for cytochrome *b*. An increased rate of cytochrome *c* oxidation leads to a transient increase in *b* reduction, driven by an increased rate of Q_p^{•-} formation.

Antimycin inhibits reduction of ubiquinone and ubisemiquinone anion by cytochrome *b*-560 at center N (steps 4*a* and 4*b* in Fig. 2). Rapid formation of Q_p^{•-} driven by cytochromes *c* + *c*₁ and iron-sulfur protein oxidation, transfers electrons to the hemes of *b*-566 and *b*-560, where they are trapped if antimycin is present (22, 24). If antimycin is absent, oxidant-induced reduction of cytochrome *b* is transient, after which cytochrome *b* is reoxidized by ubiquinol and ubisemiquinone anion at center N.

Much of the evidence for the protonmotive Q cycle was obtained by examining reduction of the cytochromes of the *bc*₁ complex under pre-steady state conditions. In these experiments ubiquinol is added to the oxidized complex, and reduction of the cytochromes is monitored in a spectrophotometer. By examining reduction of the cytochromes *b* and *c*₁ when the iron-sulfur protein is removed from the complex and by testing the effects of inhibitors on their reduction, one can deduce the Q cycle mechanism.

One unique feature of the protonmotive Q cycle is that there are two pathways for cytochrome *b* reduction (Fig. 2). Cytochrome *b* can be reduced by Q_p^{•-} at center P, as described by step 1 of Fig. 2. This is the thermodynamically favored route for *b* reduction and the one which operates when the *bc*₁ complex functions catalytically. Alternatively, under pre-steady state conditions ubiquinol or ubisemiquinone anion (Q_n^{•-}) can reduce cytochrome *b*-560 by reversal of steps 4*b* and 4*a* at center N, if the thermodynamically favored route through center P is blocked.

Two experiments most clearly demonstrate two pathways of *b* reduction. The first evidence for two pathways of *b* reduction came by examining the pre-steady state reduction of the cytochromes when the iron-sulfur protein was removed from the cytochrome *bc*₁ complex and by testing the effects of antimycin on cytochromes *b* and *c*₁ reduction in the absence and presence of iron-sulfur protein (16, 17). Removal of iron-sulfur protein from the *bc*₁ complex blocked reduction of cytochrome *c*₁ but did not block reduction of cytochrome *b*

(Fig. 2). If antimycin was added to the *bc*₁ complex, removal of the iron-sulfur protein blocked reduction of both cytochromes *b* and *c*₁ (16, 17). Reconstitution of the iron-sulfur protein to the depleted complex restored reduction of cytochrome *c*₁ and restored reduction of both cytochromes *b* and *c*₁ which had been blocked by antimycin (26).

The explanation for these results is that iron-sulfur protein is required for *c*₁ reduction through step 1 (Fig. 2), but the *b* cytochromes can be reduced by an iron-sulfur protein-independent pathway, the reversal of steps 4*b* and 4*a*, which is inhibited by antimycin. If this explanation is correct, antimycin should not inhibit reduction of either cytochrome *c*₁ or the *b* cytochromes in a pre-steady state experiment, since these should be reducible via the center P route (Fig. 2, step 1). This prediction was tested and confirmed (27). Lack of inhibition of *c*₁ reduction by antimycin in a pre-steady state experiment cannot be reconciled with linear pathways like that in Equation 2.

The second evidence for two pathways of cytochrome *b* reduction is that there are inhibitors which act specifically on one or the other of these pathways (Fig. 2). Von Jagow and co-workers (28, 29) showed that a novel class of microbial derived toxins, which include myxothiazol (28) and stigmatellin (29), inhibits cytochrome *b* reduction by the antimycin-independent pathway through center P. When added alone, myxothiazol blocks cytochrome *c*₁ reduction at step 1 (Fig. 2) but does not block cytochrome *b* reduction, which occurs by reversal of steps 4*b* and 4*a* at center N (Fig. 2) under the pre-steady state conditions of the experiment. When added together with antimycin, myxothiazol blocks reduction of both cytochromes *b* and *c*₁ (28). Myxothiazol thus mimics removal of the iron-sulfur protein from the *bc*₁ complex.

Structural analogs of ubiquinone, such as UHDBT,² also inhibit electron transfer at center P but by a different mechanism than myxothiazol. UHDBT binds preferentially to reduced iron-sulfur protein or at an interface between cytochrome *b* and reduced iron-sulfur protein (30). This increases the iron-sulfur protein midpoint potential approximately 90 mV and impedes electron transfer from the iron-sulfur cluster to cytochrome *c*₁ (30–32).

If the protonmotive Q cycle is correct, one would expect that the iron-sulfur protein is required for oxidant-induced *b* reduction, and center P inhibitors would block this reaction (Fig. 2). Iron-sulfur protein is required for oxidant-induced *b* reduction (33), and UHDBT (34) and myxothiazol (30) both inhibit the reaction.

Nuances of the Q Cycle Mechanism

One important feature of the Q cycle is that ubiquinol oxidation and ubiquinone reduction occur at two separate sites (Fig. 2). These sites, centers N and P, are topographically disposed toward the negative and positive sides of the membrane, respectively, and the ubisemiquinone ions, Q_p^{•-} and Q_n^{•-}, differ in their thermodynamic properties.

The oxidation-reduction potential of the ubiquinone/ubiquinol couple in the inner mitochondrial membrane is approximately +60 mV (35), while that of the cytochrome *b*-566 heme is typically 100–150 mV more negative (e.g. -95 mV, see above). It is frequently not understood how ubiquinol can reduce cytochrome *b*-566, since this appears to be a thermodynamically unfavorable reaction.

The oxidation-reduction potential of the ubiquinone/ubiquinol couple, *E*_m(Q/QH₂), is the arithmetic average of the oxidation-reduction potential of the ubiquinone/ubisemiqui-

² The abbreviation used is: UHDBT, 5-*n*-undecyl-6-hydroxy-4,7-dioxobenzothiazole.

none anion couple, $E_m(Q/Q^-)$, and that of the ubisemiquinone anion/ubiquinol couple, $E_m(Q^-/QH_2)$. In step 1 at center P (Fig. 2) ubiquinol transfers an electron to iron-sulfur protein, which has an oxidation-reduction potential of +280 mV (13). It follows that, if the Q^-/QH_2 couple at center P reacts at approximately +280 mV, the corresponding Q/Q^- couple attains a reduction potential as low as -160 mV. This potential is sufficiently low to reduce the low potential *b*-566 heme. In other words, exergonic oxidation of ubiquinol by iron-sulfur protein drives the otherwise unfavorable reduction of cytochrome *b*-566.

At center N cytochrome *b*-560 reduces ubiquinone to ubisemiquinone anion (step 4a in Fig. 2) and ubisemiquinone anion to ubiquinol (step 4b). The potentials of the Q/Q^- and Q^-/QH_2 couples at center N must be similar, since both are reduced by the same heme group. From the equilibrium constant for formation of ubisemiquinone anion from ubiquinone and ubiquinol and the Nernst equations for the ubiquinone/ubiquinol couple and the two ubisemiquinone anion couples, one can derive the arithmetic relationship between the oxidation-reduction potentials of the two ubisemiquinone anion couples and the concentration of ubisemiquinone anion shown in Equation 3.

$$E_m(Q/Q^-) - E_m(Q^-/QH_2) = 60 \log \frac{(Q^-)^2}{(Q)(QH_2)} \quad (3)$$

From Equation 3 it can be seen that when the potentials of the two ubisemiquinone anion couples are similar, the equilibrium constant for ubisemiquinone anion formation approaches unity, and the concentration of ubisemiquinone anion approaches the concentrations of ubiquinol and ubiquinone. This predicts that the ubisemiquinone anion at center N, Q_n^- , is relatively stable. This prediction was borne out by the detection of an appropriately stable ubisemiquinone anion in the isolated mitochondrial *bc*₁ complex (35).

At center P, where ubiquinol is divergently oxidized, the increment in potentials between the two ubisemiquinone anion couples is greater than 350 mV. Consequently, the equilibrium constant for formation of Q_p^- is less than 10^{-6} , and the predicted concentration of ubisemiquinone at center P is too low to be detected by EPR spectroscopy, under equilibrium conditions. However, De Vries and co-workers (36) detected an unstable ubisemiquinone in the mitochondrial *bc*₁ complex under non-equilibrium conditions by adding ferricyanide to rapidly oxidize the complex and adding antimycin to block reoxidation of the unstable ubisemiquinone. Since antimycin eliminates the EPR signal arising from the stable Q_n^- (35), it is likely that the ubisemiquinone species detected by De Vries and co-workers is Q_p^- .

The divergent oxidation of ubiquinol at center P requires that the ubiquinol oxidase site is close to both the iron-sulfur

cluster and the *b*-566 heme since there must be a significant kinetic advantage to electron transfer from Q_p^- to *b*-566 to overcome the 350 mV increment in potentials which otherwise favors reduction of the iron-sulfur cluster by the strongly reducing Q_p^- . Center P must also exclude water and oxygen; otherwise Q_p^- would react promiscuously. This implies that there is an anhydrous proton conduction pathway, either on cytochrome *b* or the iron-sulfur protein. How cytochrome *b* and iron-sulfur protein confer these properties on center P is one of the questions which remains to be answered to complete our understanding of energy transduction in the cytochrome *bc*₁ complex.

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